

Welcome to MIE1626



Week 1

Course goals

- Introduce standard statistical inference
- Modern data science workflow
 - Agentic coding
 - Acquire data->analyze data->present results->suggest action workflow
- Data science coding in R and Python
 - Practice coding, understand data science code
 - Understand statistical inference

Course structure

- **Tentative syllabus (not finalized until the weekend: for information only)**

https://www.cs.toronto.edu/~guerzhoy/1626s26/mie1626_syllabus.pdf

Today

- Introduction to functional programming in R
- Statistical inference

Functional programming

To make a list with the squares of the numbers in the inputs $L = [5, 6, 7]$

Define the operation “square”: $\text{square}(x) = x * x$

Start with an empty list, then add $\text{square}(5)$ to the list, then add $\text{square}(6)$, then add $\text{square}(7)$

Functional programming: write this as $\text{map}(\text{square}, L)$

- In imperative style, think about modifying variables as you go
- In functional style, write everything as if it were math
 - The computer will figure out how to actually run the code for you

Why functional programming in R?

- It runs faster in R
 - The “correct” way to write in R
- It's how most people use Pandas in Python
- It works with the statistical inference material
- It's a way to do something interesting now while making sure everyone is caught up

<https://www.cs.toronto.edu/~guerzhoy/201s20/lec/W01/lec1/lec1.html>

<https://www.cs.toronto.edu/~guerzhoy/201s20/lec/W01/lec2/lec2.html>

<https://www.cs.toronto.edu/~guerzhoy/201s20/lec/W02/lec1/lec1.html>

<https://www.cs.toronto.edu/~guerzhoy/201s20/pre/P2/p2.html>

<https://www.cs.toronto.edu/~guerzhoy/201s20/lec/W02/lec2/lec2.html>

<https://www.cs.toronto.edu/~guerzhoy/201s20/lec/W03/lec1/lec1.html>

<https://www.cs.toronto.edu/~guerzhoy/201s20/lec/W03/lec2/lec2.html>

Exercise

<https://www.cs.toronto.edu/~guerzhoy/201s20/pset/PS1/ps1.html>

Project

- Get a unique dataset
 - For example, scrape reddit
- Some kind of statistical analysis using at least one technique from the course + anything you want
- Create some kind of dashboard
- Something extra: extra analysis, testing an interesting hypothesis, . .
- Proposal:
 - ~1 page
 - Data source
 - a summary of at least 2 related papers
 - 1 or 2 people per project

Probability review

Maximum likelihood, R code

Bayesian inference intro R code

Introduction to statistical inference R code

Predictive modelling + linear and logistic regression

<https://www.cs.toronto.edu/~guerzhoy/201s20/lec/W04/lec1/predictive.pdf>

<https://www.cs.toronto.edu/~guerzhoy/201s20/lec/W04/lec1/predmodel.html>

<https://www.cs.toronto.edu/~guerzhoy/201s20/lec/W04/lec2/logistic1.html>

<https://www.cs.toronto.edu/~guerzhoy/201s20/lec/W04/lec2/logistic2.html>

BA2028

Cross-validation

https://www.cs.toronto.edu/~guerzhoy/201s20/lec/W06/variable_selection.html

Inference: take one

https://www.cs.toronto.edu/~guerzhoy/201s20/lec/W12/linreg_inference.pdf

<https://www.cs.toronto.edu/~guerzhoy/201s20/lec/W12/linreg.html>

The Truth About Linear Regression

Plan

Agentic coding/scraping

Review: frequentist inference

Multilevel models

Causal inference

Multilevel models, causal inference

<https://www.cs.toronto.edu/~guerzhoy/310f19/lec/W03/hierarchical.pdf>

<https://www.cs.toronto.edu/~guerzhoy/310f19/lec/W03/ssc2015.pdf>

<https://www.cs.toronto.edu/~guerzhoy/310f19/lec/W03/causal.pdf>